

Winter 2025: Advanced Computer Lab (CSEN 704/ DMET 706)

Course Information

Project Grade: 85%

Presentation Grade: 5%

Tasks Grade: 10% (2.5% per Task)

Deadlines:

Team Creation: 25/09/2025 11:59pm

Sprint 1: 18/10/2025 11:59pm

Sprint 2: 29/11/2025 11:59pm

Sprint 3 (extra features only): 07/12/2025 11:59pm

Please read the following instructions carefully:

- Any case of **cheating**, will result in a zero.
- It is **YOUR responsibility** to ensure that you have:
 - Read and understood everything in the description. This description contains **ALL** the information you need to know about the project.
 - Found a team to work with.
 - Submitted the correct file(s) on GitHub.
 - Shown your individual contribution on GitHub.
 - Submitted all your work for each Sprint before their deadlines.

1 Course Objectives

- Learn how to properly use the Agile Methodology to plan out a project and develop the software.
- Learn the process of following a given set of System Requirements to develop a software.
- Learn to research and master the use of the MERN Stack.
- Learn how to work together as a team on GitHub.

2 Attendance

- Cross-attendance is **NOT** allowed.
- If you must cross-attend due to any emergency, you must email/ inform the course instructor (Dr. Mervat) and if she approves, your TA will be able to give you your attendance.

3 Tasks [10%]

- You must come to the labs with the following installed on your laptops:
 - VS Code
 - Postman / (extension on VS Code)
 - Node JS
 - Git
- Starting Week 3 (Week 2 of tutorials), there will be a **graded** task required to be completed by each student **INDIVIDUALLY** each week. Therefore, students **MUST** bring their laptops with them **every lab**.
- There will be **FOUR** tasks in total.
- For every task, each student is required to **fork** the task repository (which will be posted on the CMS), **clone** the created repository, complete the task and **commit** their work to the forked repository on GitHub. Using GitHub correctly will count towards 50% (1.25%) of the task grade. The other 50% (1.25%) will be earned once your TA tests your task and can see that it works **FULLY**. No partial grades will be granted for completing part of the task. Receiving the GitHub grade is dependent on you completing the entire task in the first place.
- Note that the tasks are day-specific. Meaning a different repository with a different task for each day will be posted. Pay attention to the repository you are forking. Forking the wrong task for your tutorial day will result in a **ZERO**.

- Each student is free to use any online resources they deem fit. You cannot help each other.
- If a TA suspects that a student has copied their task from someone else, they reserve the right to test the student in the task. If the student shows little to no knowledge of how the task was done, they will receive a **ZERO** for the task.
- Failure to bring a laptop will result in receiving a grade of **ZERO** for the task. If your laptop malfunctions or you forget to bring it in, it is **YOUR** sole responsibility to borrow one and have it ready in time for the lab.
- You will be given the chance to compensate just **ONE** missed task at the end of the semester.

4 GitHub

- A GitHub repository will be created by the Product Managers for each team.
- The SCRUM Masters will be sent invitations on their **GUC student emails**.
- The SCRUM Master is required to invite all the team members to the repository. This is **solely** the SCRUM Master's responsibility.
- Every team member must have **at least THREE** meaningful commits. Examples of commits that are **NOT** meaningful include: 1 changed line of code, addition/removal of a comma or semi-colon, etc.
- Product Managers will check all commits of all team members before evaluations and reserve the right to test team members **individually** if they suspect a team member of not working or if a formal complaint was made against that team member.
- Each team must have a working version of their project on the Main (Master) branch.
- Include the University Event Management Application.xlsx file in your repository and write the name of each team member responsible for each requirement (functional or non-functional) next to each requirement.

5 Project [85%]: University Event Management Application

5.1 Overview

This project will follow the Agile Methodology; meaning it will be split into Sprints, with each Sprint lasting a set amount of time and a fully functioning version of the project with the specified System Requirements (from University_Planning_System.xlsx on the CMS) should be submitted and will be evaluated.

There will be 3 Sprints with the following lengths:

- Sprint 1: 3 weeks
- Sprint 2: 4 weeks
- Sprint 3 (presentation): 2 weeks

At the end of each Sprint, you must provide a fully working version of your software (front-end, back-end and database entries) with **ALL** of the System Requirements for that Sprint.

Please also note the following:

- You are allowed to work ahead; however, you will not earn extra points for completing any requirements for a later Sprint.
- You are allowed to push a maximum of **FIVE** requirements from Sprint 1 to Sprint 2 **only**.
- You will need to complete any unfulfilled requirements from Sprint 1 in Sprint 2 since other requirements will be dependent on the ones that have not been fulfilled.
- You are allowed to use your imagination when creating this web application. The only limitation you have is that you **MUST** use the **MERN Stack** (MongoDB, Express JS, React JS and Node JS). You may use more technologies but you may not use less.

5.2 Project Theme

The University Event Management Application is a centralized web platform designed to streamline the planning, management, and participation of campus events. It enables the **events office** to create and oversee various event types—such as bazaars, trips (Cairo or Berlin), competitions, workshops, and conferences—while ensuring proper approval workflows. **Students**, **TAs**, and **staff** can browse, register for, and track events, while doctors can propose and manage academic workshops and conferences, subject to approval. **Admins** play a validation role by verifying student/staff status and approving external company participation. **External companies and vendors** can register to join bazaars or career fairs, upload documentation, and interact with students.

- Teams must have 7-10 members. Less than 7 is not recommended because of how big the project is. More than 10 will **not be permitted**.
- Teams that choose to be less than 7 members will have to complete the full project just like the rest of the teams. No exceptions will be made.
- Teams can be cross-tutorial and cross-major

- You are each responsible for finding a team to work with. **There will be no randomization of teams.**
- Each team will have a repository on GitHub created for them under their team name and will be assigned a Product Manager (TA) to refer to when they are having problems and set appointments with during evaluations for Sprints 1 and 2. Once the teams are created, Product Managers will be assigned and their names will be posted on the CMS.
- All office hours must be requested by email to ensure giving each team a fair amount of time. Be sure to only contact your assigned Product Manager.
- Teams will be required to book appointments during **Feedback Week** with their team's Product Manager in order to receive feedback on the UI/UX of their projects.
- It is the Scrum Master's responsibility to ensure any and all tasks are divided equally and fairly among team members and that every team member does their work.
- Should conflict arise within teams, SCRUM Masters **MUST** report the issue **VIA EMAIL** to their Product Manager by the deadline set for complaints before each Sprint. Any complaints made after the complaints deadline, during or after the evaluations will not be accepted and the entire team will receive the same grade for that Sprint regardless of the issue.

ITEM	DEADLINE DATE
SPRINT 1 TEAM COMPLAINTS	13/10/2025 11:59pm
SPRINT 2 TEAM COMPLAINTS	24/11/2025 11:59pm
SPRINT 3 TEAM COMPLAINTS	04/12/2025 11:59pm

5.3 Grading Criteria and Deliverables

The grading criteria of the entire project are all outlined in the Excel sheet named **University Event Management Application.xlsx** posted on the CMS.

5.3.1 Functional Requirements

The Functional Requirements will be graded based on the following criteria:

- Works well on front-end and back-end and all changes are reflected on on the database: this earns you full credit for the requirement.
- Attempting to implement the requirement as code but it does not reflect on database and/or front-end: if it is implemented incorrectly in the back-end but it seems to be working on a front-end level, you will get partial credit.
- Not implemented at all: you get no credit.

5.3.2 Non-Functional Requirements

The Non-functional Requirements are UI (user interface) and UX (user experience), shown under the **Non-functional Requirements** tab in the in the Excel sheet named **University Event Management Application.xlsx** that will be posted on the CMS. Their evaluations will be **PURELY SUBJECTIVE** and **based on the opinion of the Product Manager assigned to each team**.

You **WILL** be graded partially on the UI/UX in Sprint 1 and partially in Sprint 2. Refer to **University Event Management Application.xlsx** for more details.

5.3.3 README.MD File [10 Marks]

Your README.MD file is ultimately a guide to how your project works. The following link to a [README.MD File Template](#) will be useful in guiding you in how to write a proper README file. It also contains the template you **must** use. Short or incomplete README.MD files will get a ZERO. **This report will be evaluated along with your Sprint 2 requirements.**

The README.MD files will be graded based on the following criteria:

5.4 Evaluations

- Evaluations will be conducted in the week following the submission date and **ONLY ON THAT WEEK**. Submission dates are **final** and will be monitored on GitHub.
- Teams will be required to pull their entire repository at the beginning of the evaluation to ensure only the latest version of their project on GitHub is being evaluated.
- SCRUM Masters must book an appointment with their Product Manager in order to evaluate. Product Manager will announce when appointments are available via email.
- **ALL team members MUST attend the evaluations.** Missing team members/teams who do not show up for any of the evaluations will receive a **ZERO for that entire Sprint**. This is nonredeemable and no excuses will be accepted unless under very specific circumstances and emergencies **with proof** which must be delivered to Dr. Mervat.

6 Presentation[5%]: University Event Management Application Extra Features

- Each team will be required to **come up with and fully implement FIVE extra features** for their application. **Only the extra features** are due on the deadline specified for Sprint 3. Teams do not need to upload their presentations onto GitHub on this deadline.

README.MD file	Marks	Comments
Format is NOT .txt or PDF	0.25	–
Project Title	0.5	–
Motivation	0.5	–
Build Status	1	All issues with your project must be mentioned here. There is no such thing as a perfect project.
Code Style	0.5	Includes how the which naming convention was used, etc..
Screenshots	1.5	Can be screenshots (pictures) or a video of how system looks/works. If screenshots of the system are used, include at least 5 screenshots.
Tech/Framework used	0.5	Explain what technologies you used (MERN stack and what else if anything)
Features	1	List all the features on the system. Since this is due in Sprint 2, you do not need to include the new features.
Code Examples	1	Add examples of your code here. Add at least 5 snippets of functions.
Installation	0.5	What needs to be installed to run your project?
API References	1	At least 5 routes need to be mentioned.
Tests	1	Testing used in Postman (screenshots of Postman tests) is fine. JEST is also accepted. Add at least 5 tests.
Contribute	0.25	How can others contribute to your project? Writing that the project is perfect is not acceptable.
Credits	0.25	This includes any source that you used online for help INCLUDING YouTube videos
License	0.25	For example: If Stripe is used, Apache 2.0 license Must be included in this section.

- Each team will be required to demo their extra features in a presentation and explain the importance and impact of the features they chose to add to their project.
- SCRUM Masters will need to book appointments for their teams to present. The Product Managers will announce when appointments are available via email.
- Any **THREE** team members will be chosen by the TAs **right before the presentation** to give the presentation. This means **ALL** team members must be ready and must understand the new features and how their project works, and be ready to answer questions.
- **Several of the course TAs** will be present during the presentations and will weigh in on the usefulness and execution of the five extra features produced by each team member. A total grade for the presentation will be agreed upon by all the present TAs.

- Presentation grading will also include presentability of the 3 presenters themselves.
- The full presentation grading breakdown will be posted by the Sprint 3 deadline.